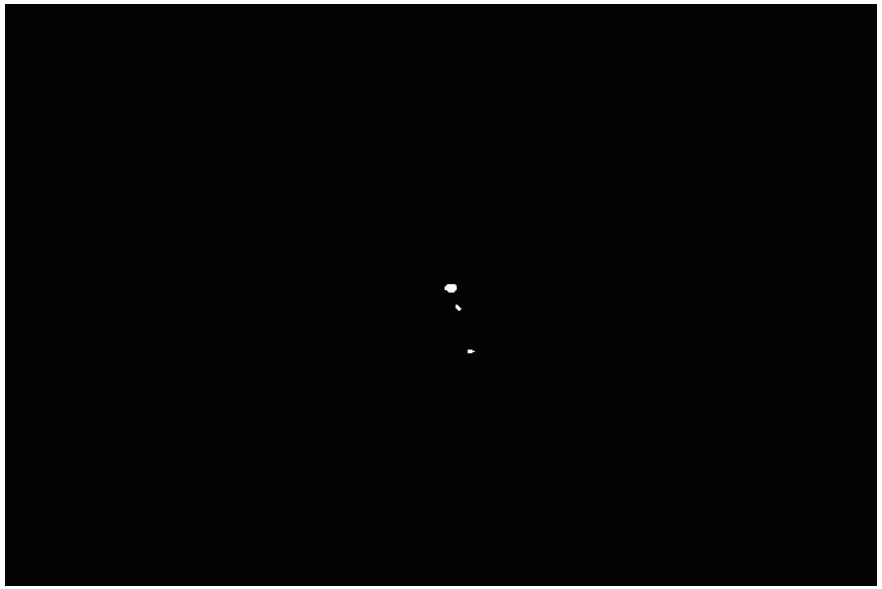
数据标注

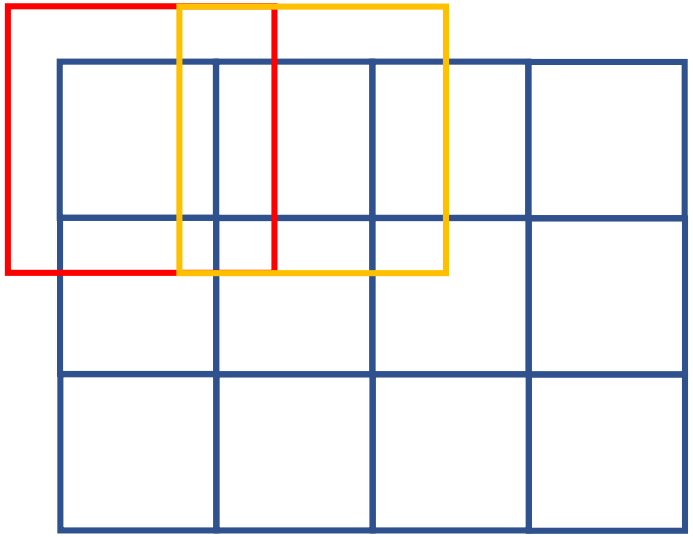
链接：https://pan.baidu.com/s/1ARPFvXDnvTNOi3B6UV1xJA 提取码：vsv1

- change_mean_std.py
- compute_mean_std.py
- datamark_all.zip
- datamark_mean_std.zip

项目进展

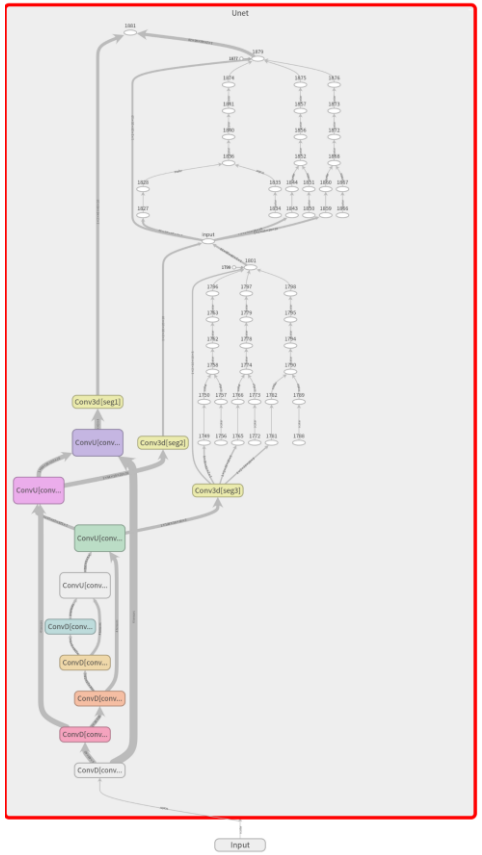
对于图像合龙之后存在边框的问题。由于脑肿瘤分割的边缘为黑色，所以可以采用padding实现。

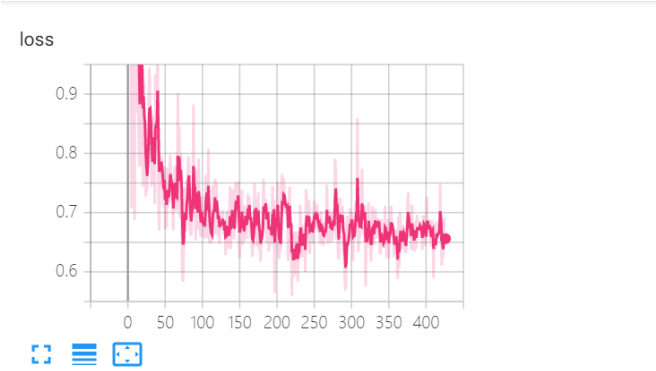




解决思路：
将图像原始尺寸为500*400*N，由于Unet输入为4的倍数。
所以将原始尺寸分别裁剪为500*400*60,100*100*40,20*20*20（假设N=120）。
多尺度预测，再拼起来。取概率大的。

网络结构

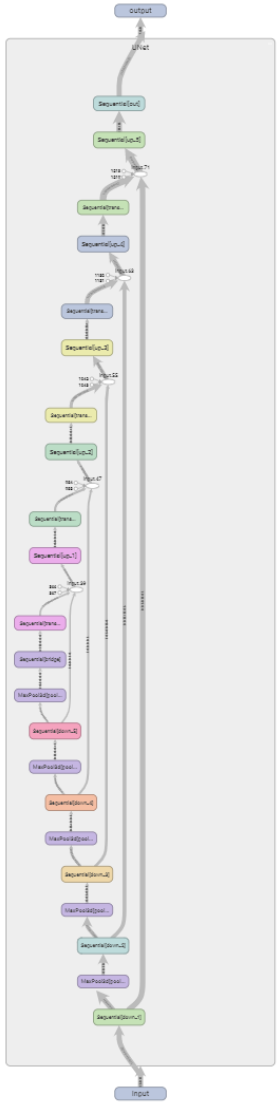




300代

400代

新的网络结构



遇到问题：
预测尺寸报错64*64*32/128*128*64

Deep Smoke Segmentation

- 创新点1：由于自己标注的不准确性，采用自动生成烟雾与背景图叠加。

Fig. 1 shows several pure smoke samples. Each pure smoke sample is a 256×256 RGBA image containing 4 channels, i.e., three RGB color channels ($\mathbf{S}$) and one alpha channel ($\boldsymbol{\alpha}$). We use the linear color composition of a pure smoke image ($\mathbf{S}$ and $\boldsymbol{\alpha}$) and a background RGB image $\mathbf{B}$ to generate a new smoke RGB image $\mathbf{I}$, expressed as:

$$\mathbf{I}(\boldsymbol{x}) = (1 - \alpha(\boldsymbol{x}))\mathbf{B}(\boldsymbol{x}) + \alpha(\boldsymbol{x})\mathbf{S}(\boldsymbol{x}) \tag{1}$$

Fig. 1. Pure smoke images generated by computer graphics.

To make the training samples diverse, we used the Places365-Standard dataset [52] as background images. In the synthesis process, we randomly select a background image from the Places365-Standard dataset and a pure smoke image from the PureSmoke dataset, then combine the two images to generate a synthesized smoke image. Pure smoke images in Fig. 1 have similar gray color. To further augment smoke images, we change the alpha values of pure smoke images to change the concentrations of smoke images by a parameter β, expressed as:

$$I_R(\boldsymbol{x}) = (1 - \alpha(\boldsymbol{x}))B_R(\boldsymbol{x}) + \alpha(\boldsymbol{x})\beta S_R(\boldsymbol{x}) \tag{2}$$

$$I_G(\boldsymbol{x}) = (1 - \alpha(\boldsymbol{x}))B_G(\boldsymbol{x}) + \alpha(\boldsymbol{x})\beta S_G(\boldsymbol{x}) \tag{3}$$

$$I_B(\boldsymbol{x}) = (1 - \alpha(\boldsymbol{x}))B_B(\boldsymbol{x}) + \alpha(\boldsymbol{x})\beta S_B(\boldsymbol{x}) \tag{4}$$

where β is a global coefficient with range (0, 1) used to control the color of smoke, I_R, I_G and I_B are the red, green and blue channels

Fig. 2. Pure smoke images and synthesized images.

- 创新点2：利用两路unet的结构，深层获得全局语义信息，浅层获得细节信息

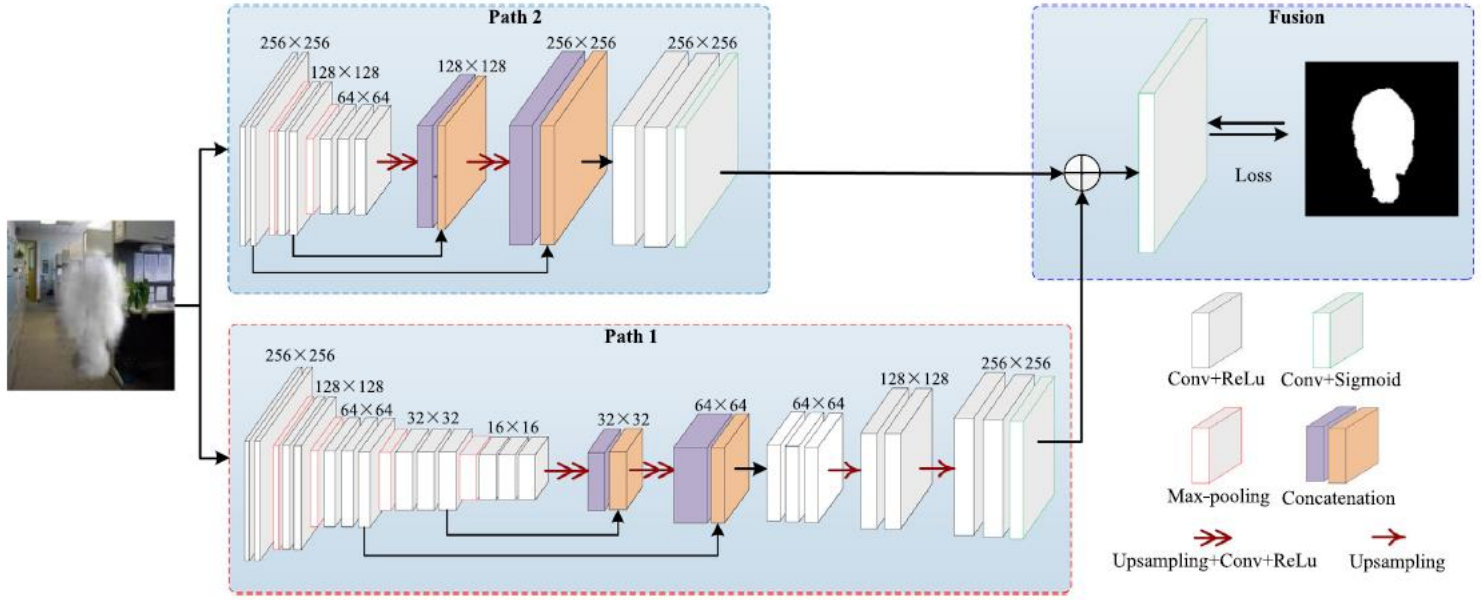
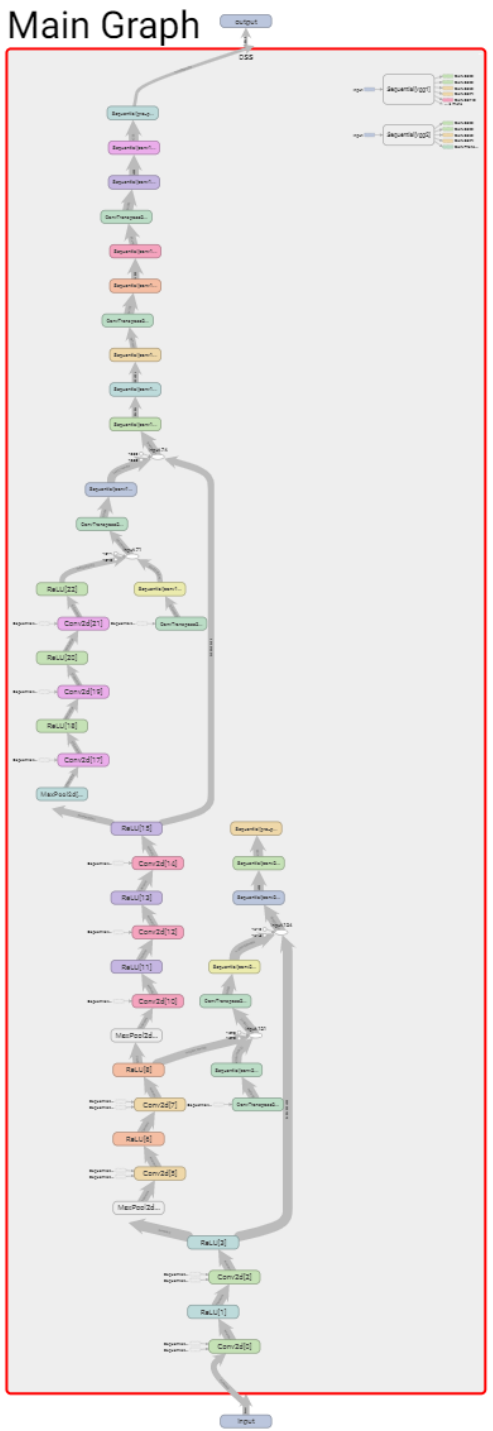


Fig. 3. Our network structure with two paths and skip layers.



评价指标

$$\mathrm{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathbf{P}_i \cap \mathbf{G}_i}{\mathbf{P}_i \cup \mathbf{G}_i}$$

$$\mathrm{Mse}_i = \frac{\sum_{k=1}^{h_i \times w_i} \left\| p_i^k - g_i^k \right\|^2}{h_i \times w_i} \qquad \mathrm{mMse} = \frac{1}{N}\sum_{i=1}^{N}\mathrm{Mse}_i$$

Fig. 4. Synthetic smoke images and corresponding ground truths of alpha channels from (a) DS01, (b) DS02 and (c) DS03.

Comparisons on DS01.

Methods	mIoU(%)	mMse
FCN-8 s [16]	64.03	0.3221
SegNet [38]	56.94	0.3976
Static Map Detection [19]	62.88	0.3209
Text-Block FCN [40]	66.67	0.3021
Deeplab v1 [49]	68.41	0.2981
LRN [64]	66.43	0.3069
Our method	**71.04**	**0.2745**

Comparisons on DS02.

Methods	mIoU(%)	mMse
FCN-8 s [16]	63.28	0.3353
SegNet [38]	56.77	0.4173
Static Map Detection [19]	61.50	0.3379
Text-Block FCN [40]	65.85	0.3196
Deeplab 1 [49]	68.97	0.3030
LRN [64]	67.71	0.3078
Our method	**70.01**	**0.2894**

Comparisons on DS03.

Methods	mIoU(%)	mMse
FCN-8 s [16]	64.38	0.3202
SegNet [38]	57.18	0.4047
Static Map Detection [19]	62.09	0.3255
Text-Block FCN [40]	66.20	0.3070
Deeplab v1 [49]	68.71	0.3010
LRN [64]	67.46	0.3041
Our method	**69.81**	**0.2861**

实验结果

对于生成的烟雾图片

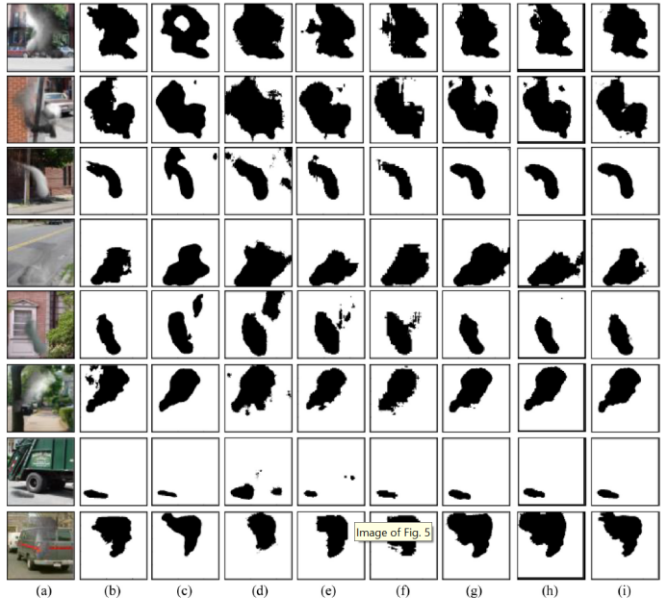


Fig. 5. Results of synthetic data. (a) Synthetic data. (b) Ground truths. (c) FCN. (d) SegNet. (e) SMD. (f) TBFCN. (g) Deeplab v1. (h) LRN. (i) Our method.

对于真实的的烟雾图片

Fig. 6. Segmentation results of real smoke images. (a) Real images. (b) FCN. (c) SegNet. (d) SMD. (e) TBFCN. (f) Deeplab v1. (g)LRN. (h) Our method.

FP

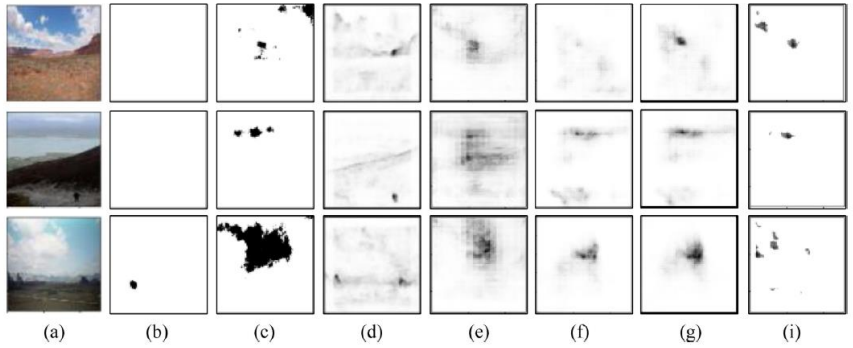


Fig. 7. Some examples of false positives. (a) Real images. (b) FCN. (c) SegNet. (d) SMD. (e) TBFCN. (f) Deeplab v1. (g) LRN. (i) Our method.

视频分割结果

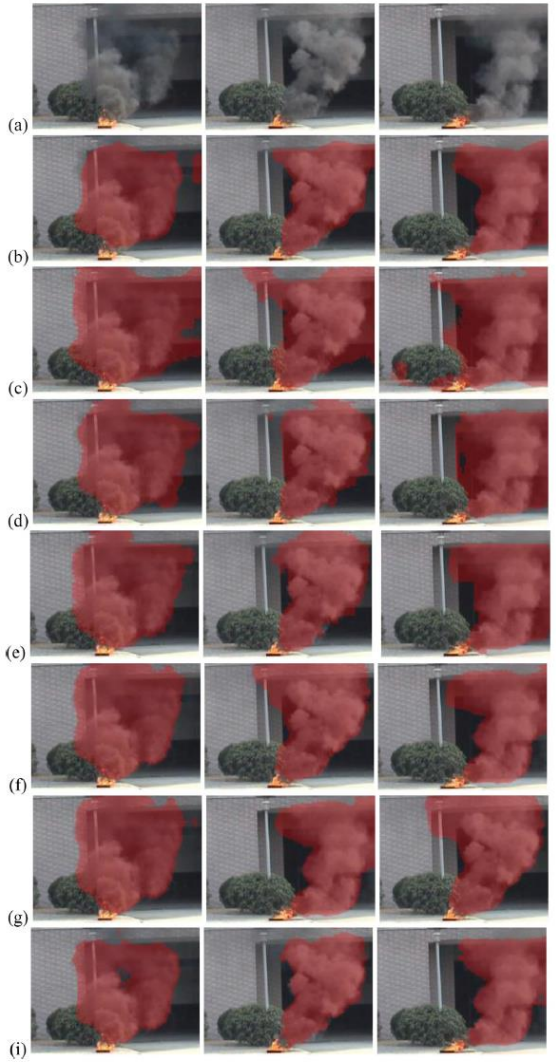


Fig. 9. Segmentation results on the black smoke video. (a) Original frames. Overlapped results of (b) FCN, (c) SegNet, (d) SMD, (e) TBFCN, (f) Deeplab v1, (g) LRN and (i) Our method.

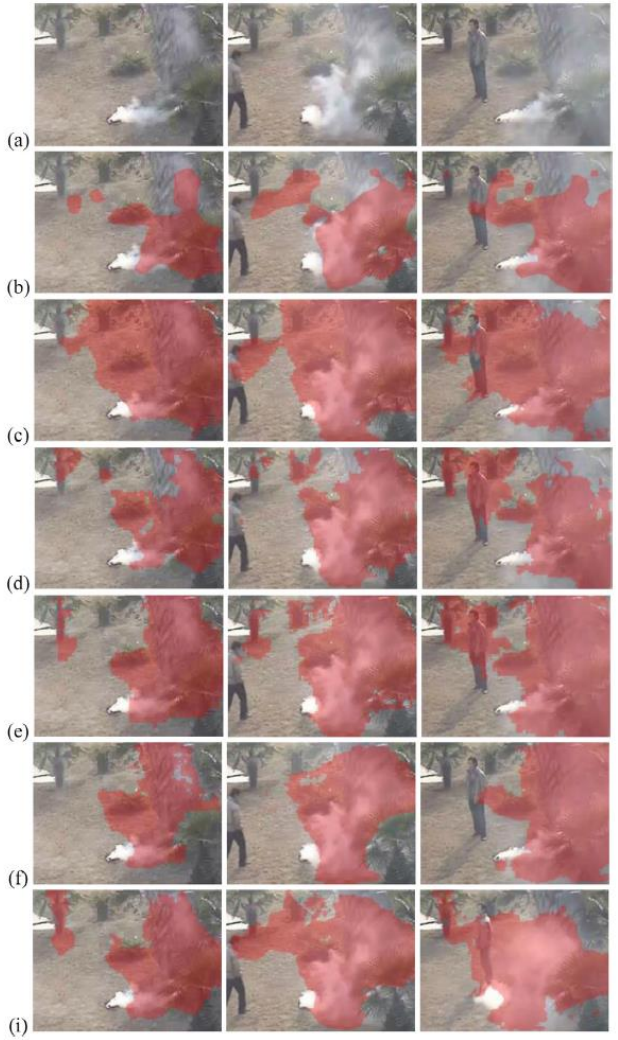


Fig. 10. Segmentation results on the white smoke video. (a) Original frames. Overlapped results of (b) FCN, (c) SegNet, (d) SMD, (e) TBFCN, (f) Deeplab v1, (g) LRN and (i) Our method.